Learning in the Open: Integrating Open Source Contributions into the Classroom

Ryan C. Cooper, Inessa Pawson,
Ryan Curtin, Mohammad Mundiwala

SciPy 2026

How can we make open source a first-class learning experience, rather than something students discover by accident?

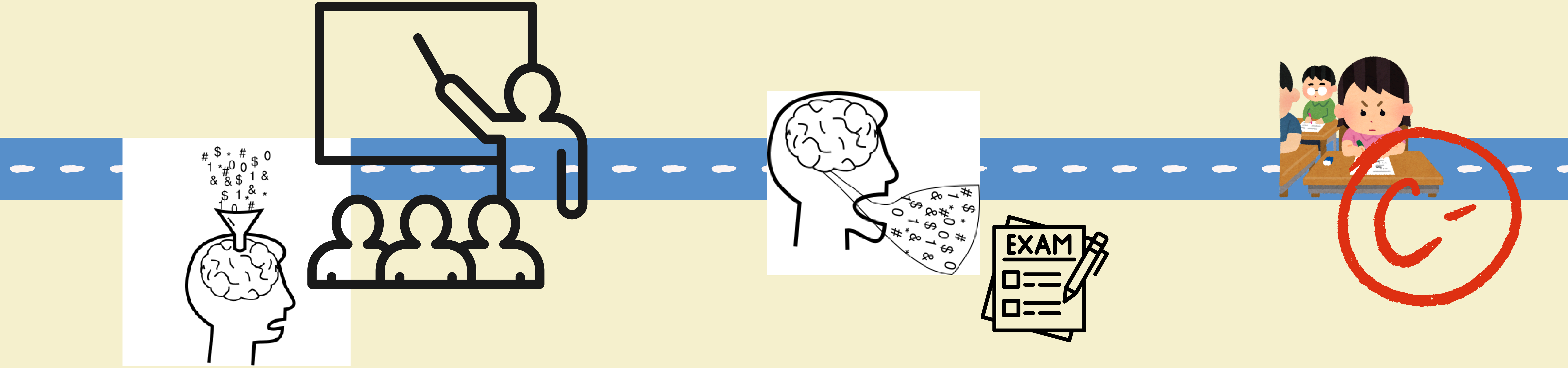
Introducing the Open Source Experience

Students participate in scientific software development to learn skills

Learning outcome goals:

- Build transferable skills
- Communicate technical information
- Develop new open source contributors

Traditional Learning and Assessment



*[Schinske and Tanner 2017]

Traditional Outcome

36

6

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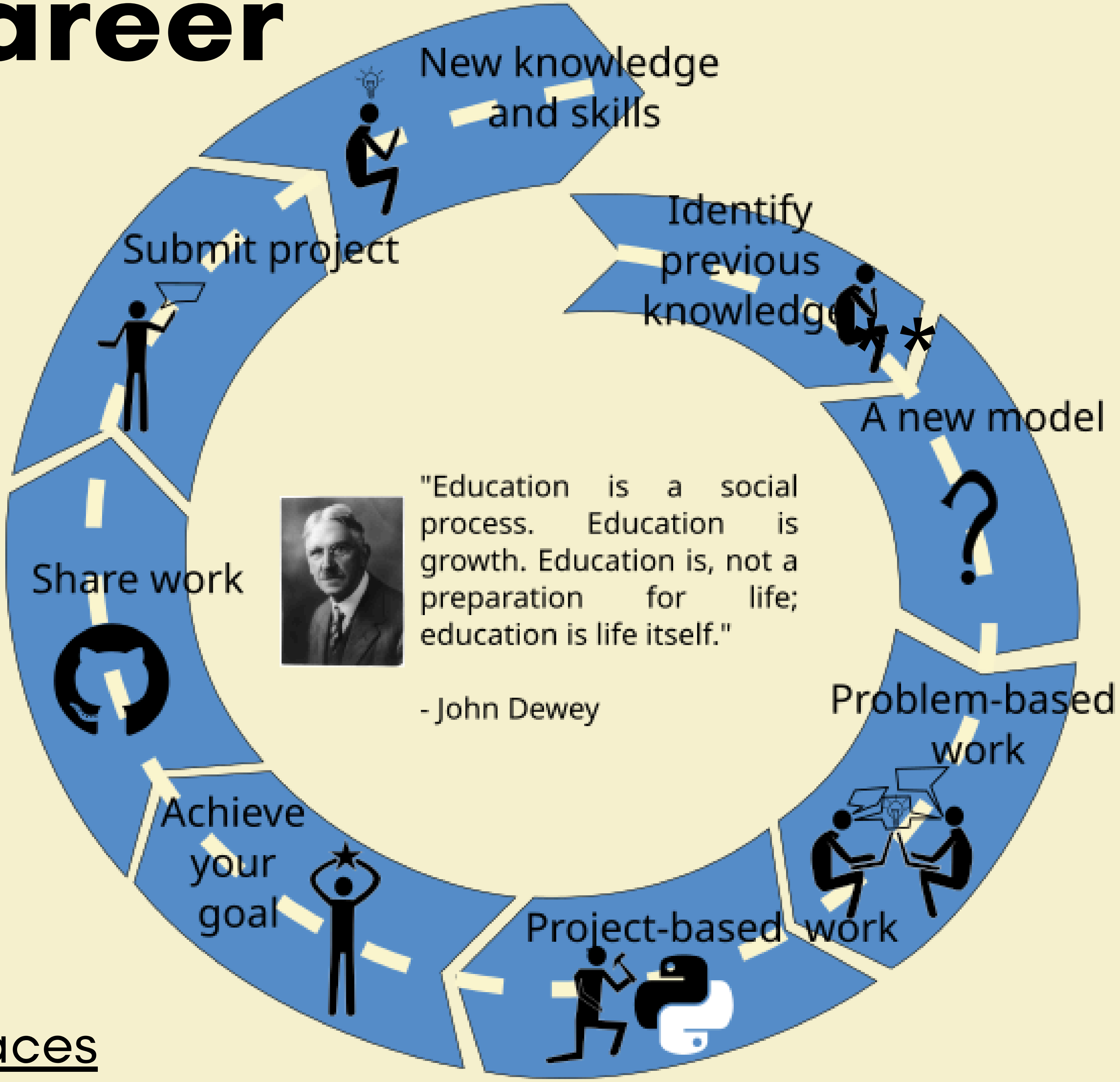
720

Learning for Your Career

Learning requires a genuine experience*

*Dewey 1929, The Quest for Certainty

**Campbell 1949, Hero with a Thousand Faces



Class Planning and Experience

Identify collaborators

- 6 NumFocus projects

Onboard maintainers

- Define deliverables
- Set responsibilities and schedule for mentorship

Onboard students

- Pick a project
- Decide on deliverables
- Connect with the mentor

Class experience

- Attend community meetings
- Meet regularly with the mentor
- Work independently on deliverables
- Course hosted 7 student-led updates
- Students ended course with exit interview

Project Collaborations

julia

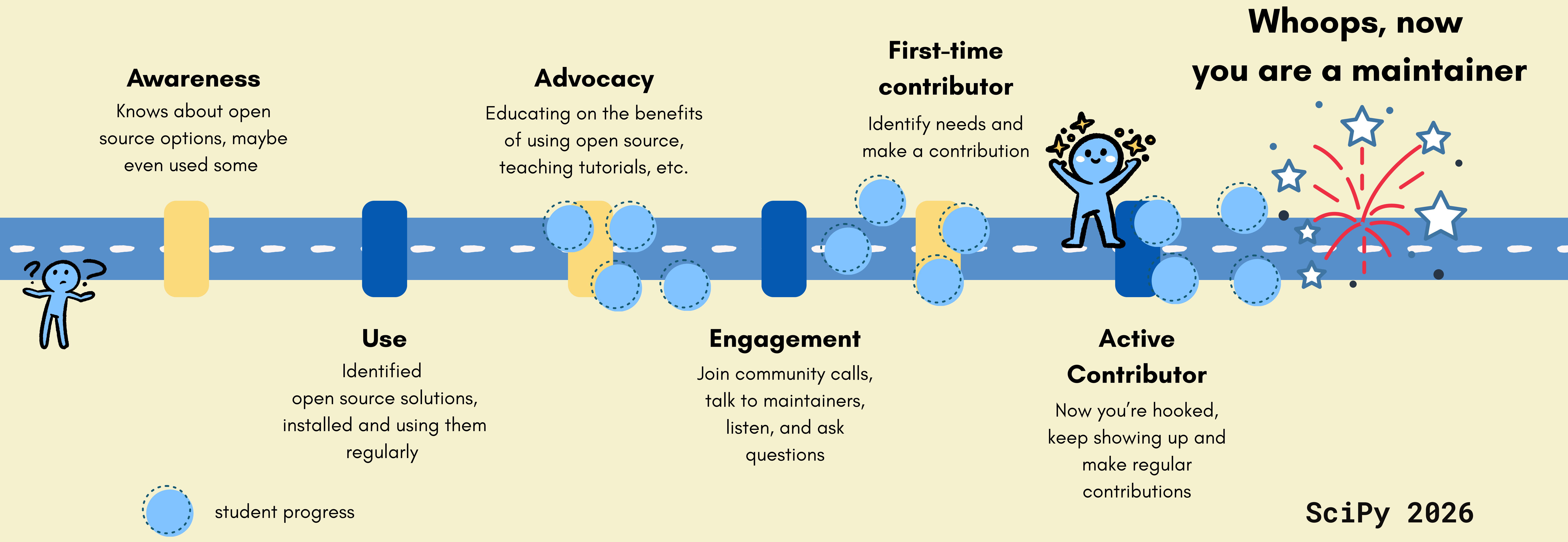
BLOSC

MNE
MEG + EEG ANALYSIS & VISUALIZATION

mlpack

BiocPy

Contributor Pipeline



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Case Study: Mohammad Mundiwala + mlpack

Meet the hero 1
Mohammad is grad student working on sustaninable manufacturing, not a coder.

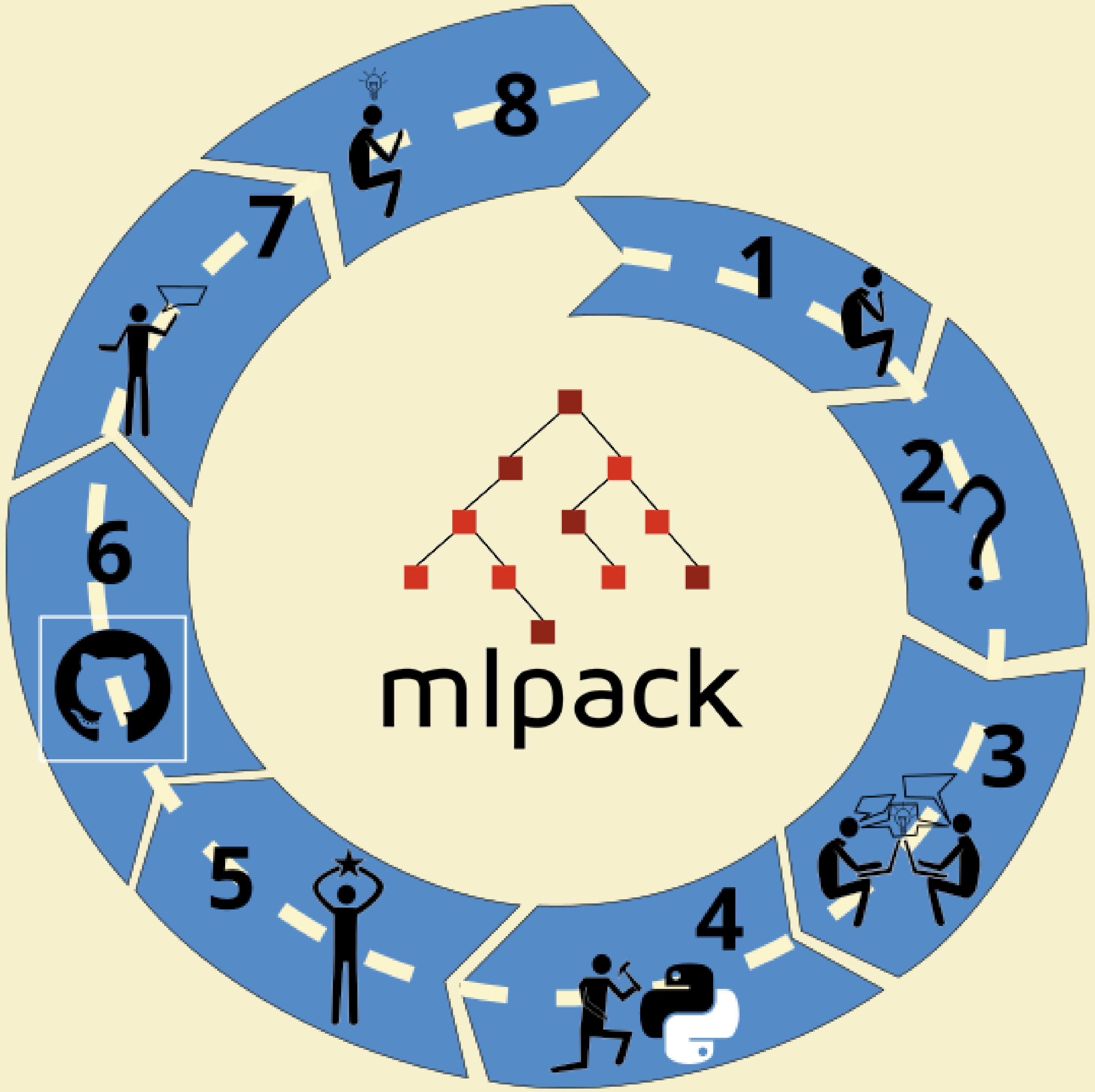
The call 2
mlpack wanted to extend C++ machine learning functions

Its dangerous to go alone 3
Mentorship, env setup, and community connection

The trials 4
Translate Empirical Mode Decomposition (EMD) into C++

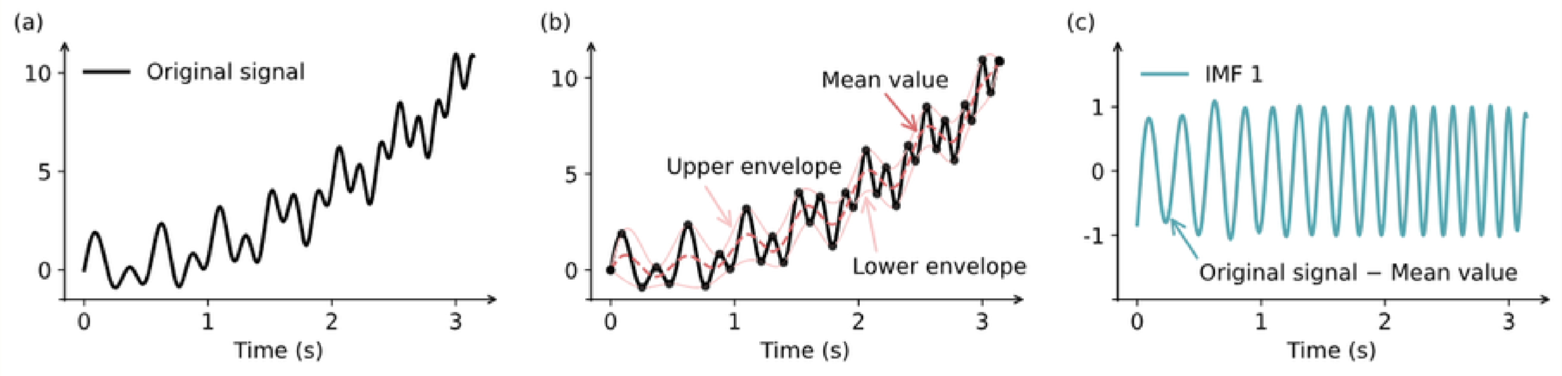
Save the princess 5
Code works! This could be a new feature!

Return of the hero 6
Mo documents process, creates tutorials, and prepares PR



Case Study: Mohammad Mundiwala + mlpack

Use the force - PR review



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Next Adventure awaits

- EMD merged into MLPack
- New tools and collaborators
- FOSS work included in next grant proposal
- mlpack ready for new applications in manufacturing and engineering

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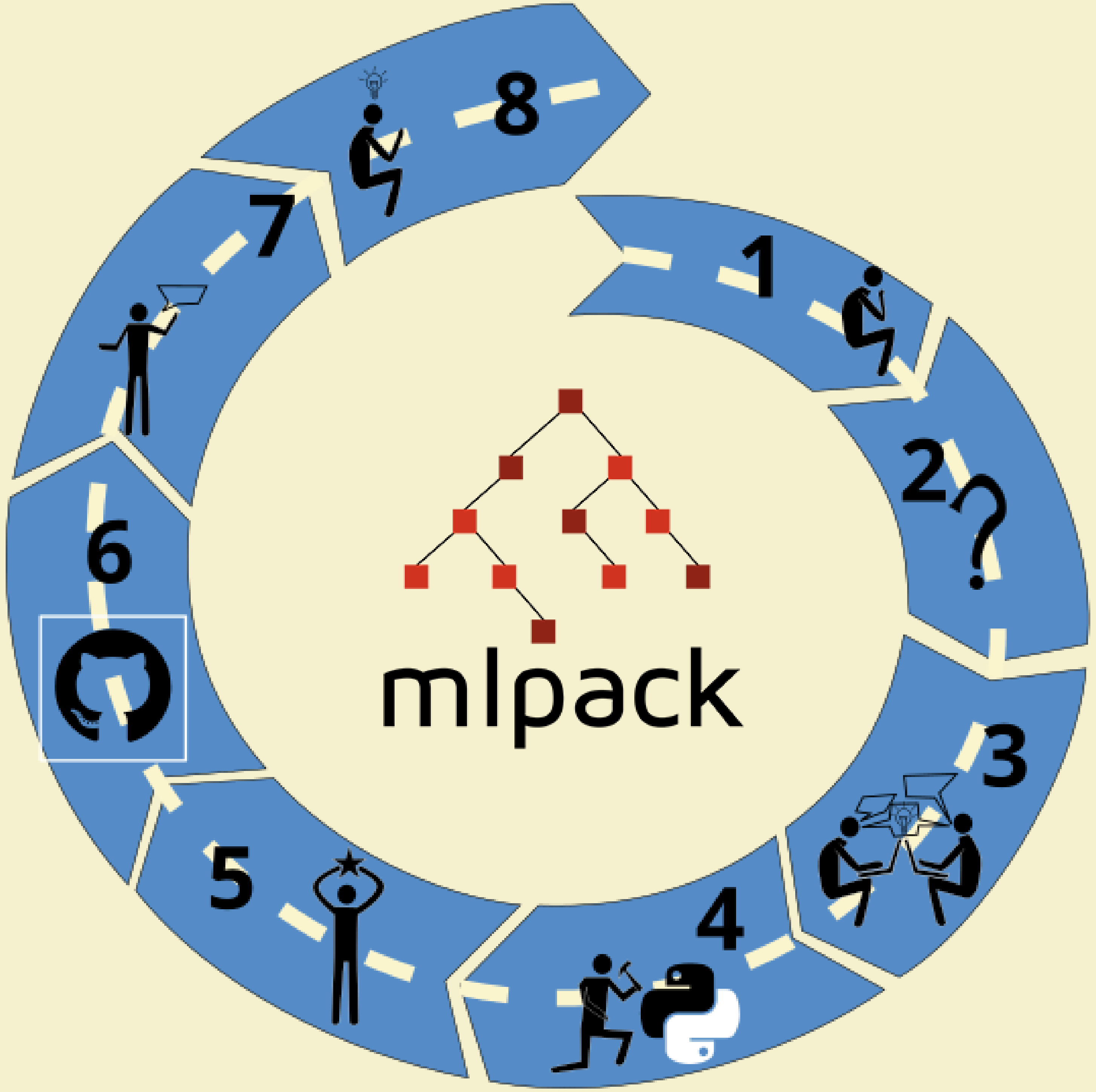

Open Source Experience Outcomes

- Achieved all learning outcomes
- Every student accomplished a learning journey
- These are the highlights from closeout interviews

Open Source
Experience Outcomes

Communicate Technical Information

- 100% updates on technical details
- 66% completed multiple tutorials and shated lessons
- 58% went through one Pull Request process
- 33% made tutorials
- 33% had multiple PRs

Open Source Experience Outcomes

Build Transferrable Skills

- 66% identified career skills
- 66% discussed GitHub lessons
- 58% connected the development to ongoing research
- 33% took on a new topic
- 33% addressed imposter syndrome

Open Source
Experience Outcomes

Develop Open Source Contributors

- 75% stressed that time was an issue
- 66% identified future project goals and opportunities
- 50% specifically mentioned mentor as role model
- 33% delievered tutorials to new users

Lessons Learned

Create structure for new contributors

- structure matters for onboarding
- every new contributor is on their own learning journey
- start small

Community matters

- Commodity code and AI bots causing burnout
- Need events and

Encourage *all* Contributions

- Community calls
- Journal notes on installation
- Guided group discussions work great

Align goals and contributions

- More motivation
- Stay connected with vested interest

Thanks so much to our students, maintainers (mentors), and University of Connecticut